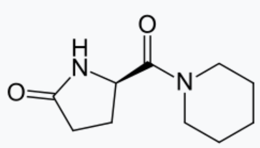
Fasoracetam

Fasoracetam is a research chemical of the racetam family.[3] It is a putative nootropic that failed to show sufficient efficacy in clinical trials for vascular dementia. It is currently being studied for its potential use for attention deficit hyperactivity disorder.[2][4]

Fasoracetam



Names

IUPAC name

(5*R*)-5-(Piperidine-1-carbonyl)pyrrolidin-2-one

Other names	(5*R*)-5-Oxo-D-prolinepiperidinamide monohydrate, NS-105, AEVI-001, LAM 105, MDGN-001, NFC 1[1][2]

Identifiers

CAS Number	110958-19-5 (https://commonchemistry.cas.org/detail?cas_rn=110958-19-5) ✓
3D model (JSmol)	Interactive image (htt

	ps://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DC%28N1CCCCC1%29%5BC%40%40H%5D2NC%28%3DO%29CC2)
ChEMBL	ChEMBL2106179 (https://www.ebi.ac.uk/chembldb/index.php/compound/inspect/ChEMBL2106179) ✗
ChemSpider	171980 (https://www.chemspider.com/Chemical-

	Structure.171980.html)
KEGG	C13311 (https://www.kegg.jp/entry/C13311)
PubChem CID	198695 (https://pubchem.ncbi.nlm.nih.gov/compound/198695)
UNII	42O8UF5CJB (https://precision.fda.gov/uniisearch/srs/unii/42O8UF5CJB)
CompTox Dashboard	DTXSID9048855

(EPA)	(https://comptox.epa.gov/dashboard/chemical/details/DTXSID9048855)

InChI

InChI=1S/C10H16N2O2/c13-9-5-4-8(11-9)10(14)12-6-2-1-3-7-12/h8H,1-7H2,(H,11,13)/t8-/m1/s1 ✓
Key: GOWRRBABHQUJMX-MRVPVSSYSA-N ✓

InChI=1/C10H16N2O2/c13-9-5-4-8(11-9)10(14)12-6-2-1-3-7-12/h8H,1-7H2,(H,11,13)/t8-/m1/s1
Key: GOWRRBABHQUJMX-MRVPVSSYBL

SMILES

O=C(N1CCCCC1)[C@@H]2NC(=O)CC2

Properties	
Chemical formula	$C_{10}H_{16}N_2O_2$
Molar mass	196.250 g·mol^{-1}

Pharmacology	
Routes of administration	Oral
Legal status	AU: S4 (Prescription only) US: Not FDA approved

Except where otherwise noted, data are given for materials in their data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).

✗ verify (https://en.wikipedia.org/w/index.php?title=Special:ComparePages&rev1=444354364&page2=Fasoracetam) (what is ✓✗ ?)

Infobox references

Fasoracetam appears to agonize all three groups of metabotropic glutamate receptors and has improved cognitive function in rodent studies.[5] It is orally bioavailable and is excreted mostly unchanged via the urine.[6]

Fasoracetam was discovered by scientists at the Japanese pharmaceutical company Nippon Shinyaku, which brought it through Phase 3 clinical trials for vascular dementia, and abandoned it due to lack of efficacy.[5][7]

Scientists at Children's Hospital of Philadelphia led by Hakon Hakonarson have studied fasoracetam's potential use in attention deficit hyperactivity disorder.[5] Hakonarson's company neuroFix tried to bring the drug to market for this use; neuroFix acquired Nippon Shinyaku's clinical data as part of its efforts.[7][8] neuroFix was acquired by Medgenics in 2015.[8] Medgenics changed its name to Aevi Genomic Medicine in 2016.[9]

Clinical trials in adolescents with ADHD who also have mGluR mutations started in 2016.[8] While Fasoracetam may be effective in the treatment of ADHD in people with specific mGluR mutations, these represent around 10% of total ADHD cases, and Fasoracetam is likely ineffective in all other cases.[10][11] Studies showing improvements in cognitive function from fasoracetam have exclusively been done on rodents.[10]

Legality

Australia

Fasoracetam is a schedule 4 substance in Australia under the Poisons Standard (February 2020).[12] A schedule 4 substance is classified as "Prescription Only Medicine, or Prescription Animal Remedy – Substances, the use or supply of which should be by or on the order of persons permitted by State or Territory legislation to prescribe and should be available from a pharmacist on prescription."[12]

See also

- Drug repositioning

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